US012413605B2

(12) **United States Patent**
Saito et al.

(10) **Patent No.: US 12,413,605 B2**
(45) **Date of Patent: Sep. 9, 2025**

(54) **SIGNAL DETECTION APPARATUS, VEHICLE, AND METHOD**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 82 days.

(21) Appl. No.: **18/341,771**

(22) Filed: **Jun. 27, 2023**

(65) **Prior Publication Data**

US 2024/0007486 A1 Jan. 4, 2024

(30) **Foreign Application Priority Data**

Jul. 1, 2022 (JP) 2022-107261

(51) **Int. Cl.**
H04L 9/40 (2022.01)
H04L 67/12 (2022.01)

(52) **U.S. Cl.**
CPC ***H04L 63/1416*** (2013.01); ***H04L 63/1483*** (2013.01); *H04L 67/12* (2013.01)

(58) **Field of Classification Search**
CPC . H04L 63/1416; H04L 63/1483; H04L 67/12; H04W 12/121
See application file for complete search history.

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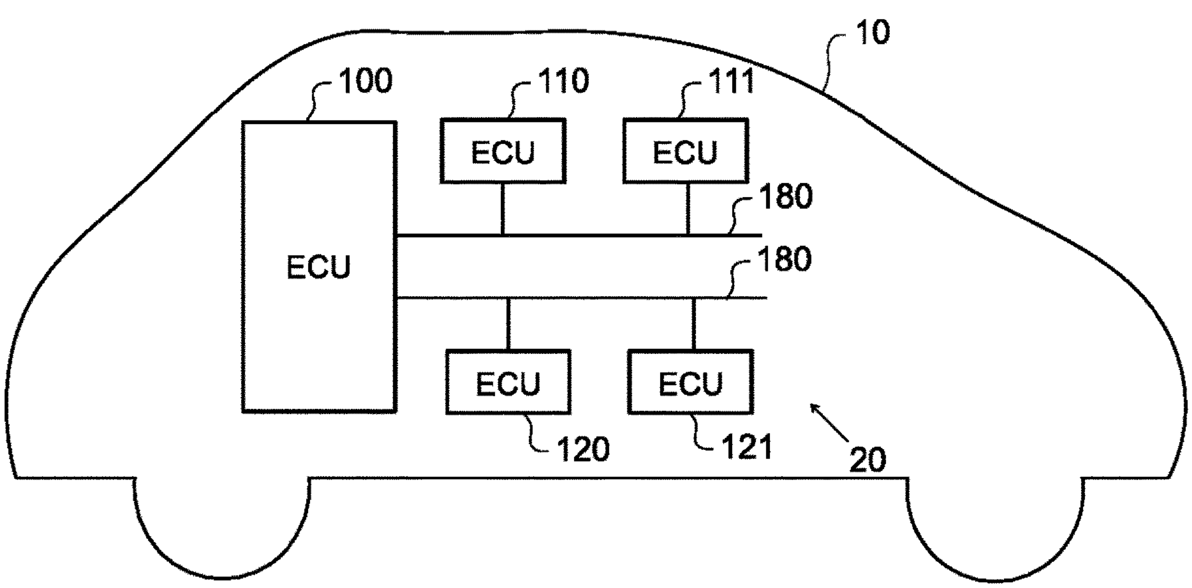
Primary Examiner — Javier O Guzman

(57) **ABSTRACT**

A signal detection apparatus includes a signal detection unit which detects signals expected to be input to a communication network in a predetermined cycle, and a determination unit which determines whether detection timing of a second signal detected by the signal detection unit after a first signal which is set as a reference signal for determining an illegal signal is in a predetermined range within which reference timing falls among the signals detected by the signal detection unit, and whether a shift between the detection timing of the second signal and the reference timing is a first threshold or less, in which the determination unit sets the second signal as the reference signal when the shift is the first threshold or less, and sets the first signal as the reference signal without setting the second signal as the reference signal when the shift exceeds the first threshold.

7 Claims, 6 Drawing Sheets



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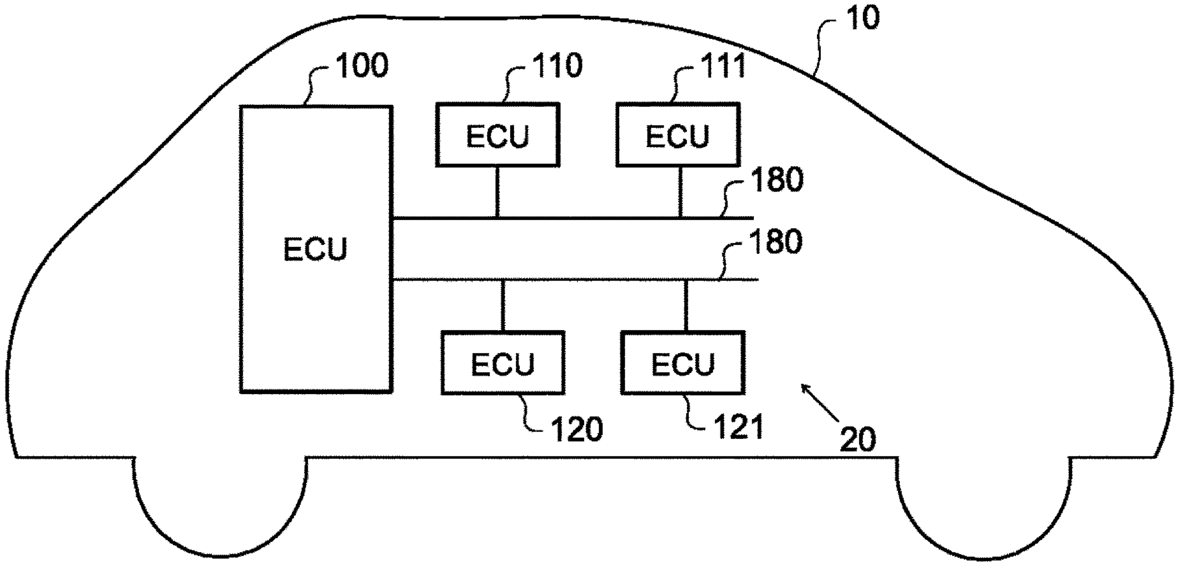

FIG.1

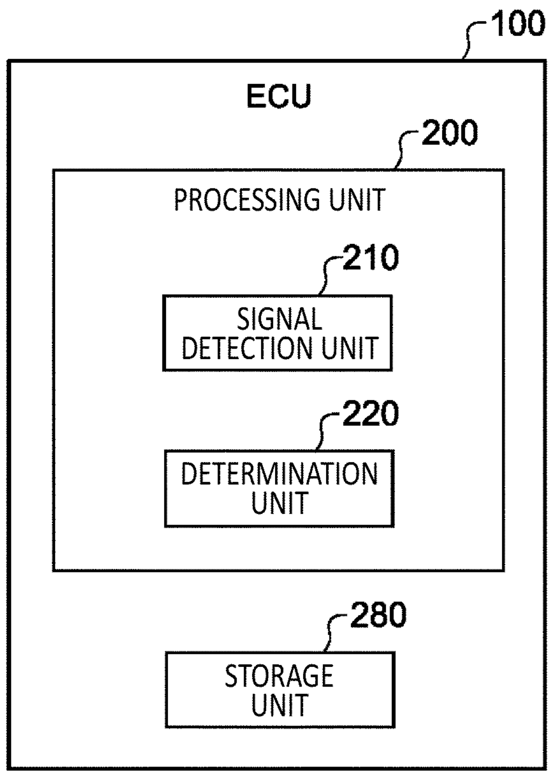
100
ECU
200
PROCESSING UNIT
210
SIGNAL
DETECTION UNIT
220
DETERMINATION
UNIT
280
STORAGE
UNIT

FIG.2

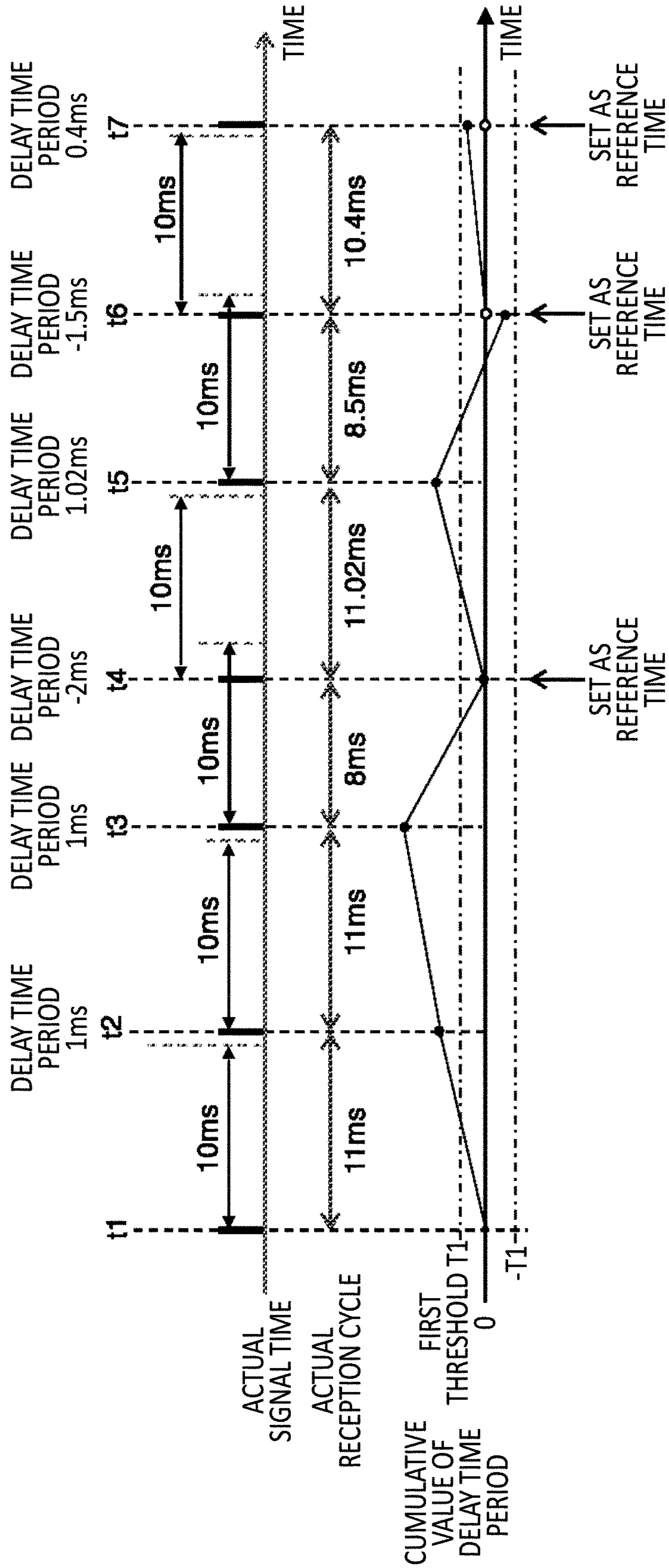
DELAY TIME PERIOD 1ms
DELAY TIME PERIOD 1ms
DELAY TIME PERIOD -2ms
DELAY TIME PERIOD 1.02ms
DELAY TIME PERIOD -1.5ms
DELAY TIME PERIOD 0.4ms
t1
t2
t3
t4
t5
t6
t7
10ms
10ms
10ms
10ms
10ms
10ms
TIME
ACTUAL SIGNAL TIME
ACTUAL RECEPTION CYCLE
11ms
11ms
8ms
11.02ms
8.5ms
10.4ms
CUMULATIVE VALUE OF DELAY TIME PERIOD
FIRST THRESHOLD T1
0
-T1
TIME
SET AS REFERENCE TIME
SET AS REFERENCE TIME
SET AS REFERENCE TIME

FIG.3

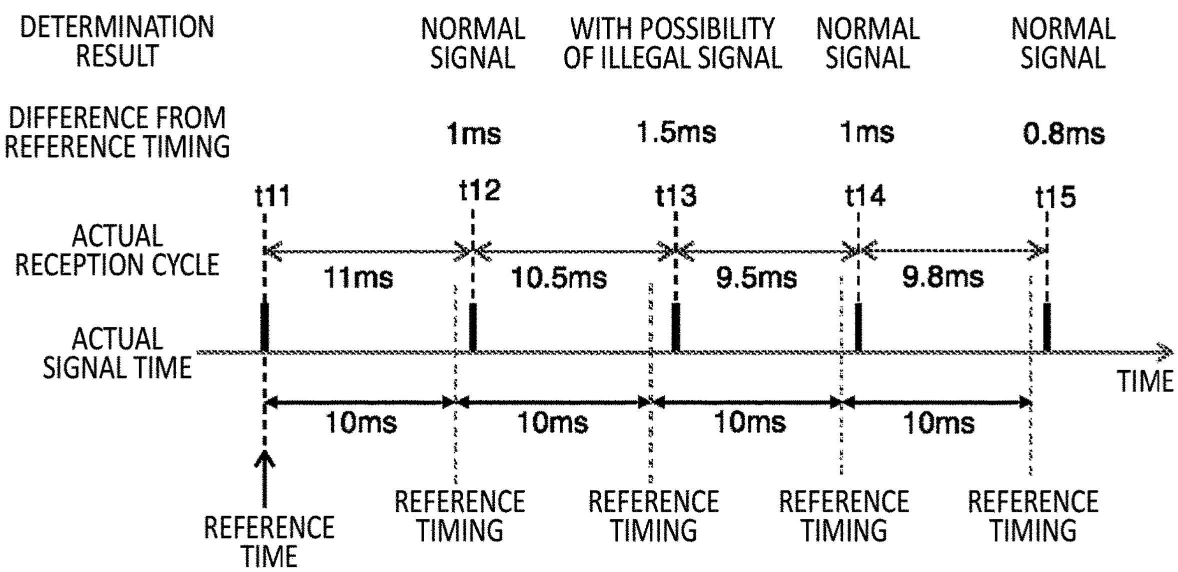
DETERMINATION RESULT
NORMAL SIGNAL
WITH POSSIBILITY OF ILLEGAL SIGNAL
NORMAL SIGNAL
NORMAL SIGNAL
DIFFERENCE FROM REFERENCE TIMING
1ms
1.5ms
1ms
0.8ms
t11
t12
t13
t14
t15
ACTUAL RECEPTION CYCLE
11ms
10.5ms
9.5ms
9.8ms
ACTUAL SIGNAL TIME
TIME
10ms
10ms
10ms
10ms
REFERENCE TIME
REFERENCE TIMING
REFERENCE TIMING
REFERENCE TIMING
REFERENCE TIMING

FIG.4

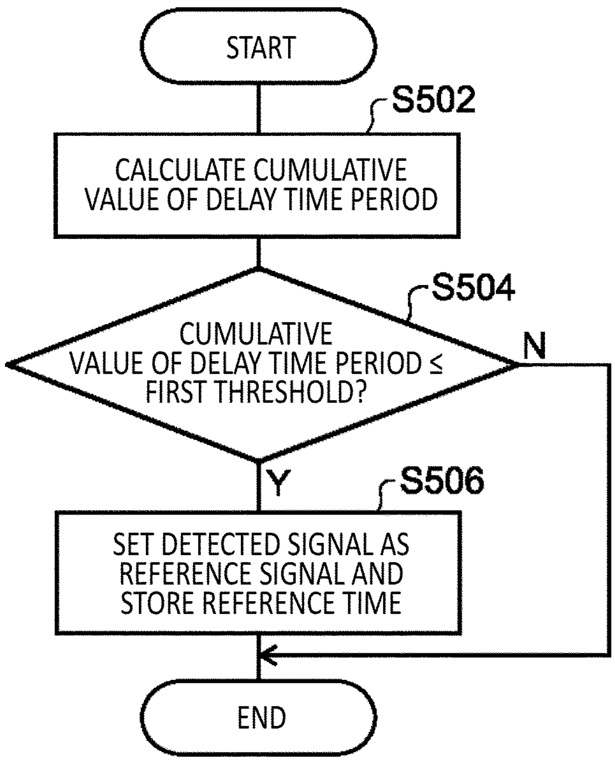
START
S502
CALCULATE CUMULATIVE VALUE OF DELAY TIME PERIOD
S504
CUMULATIVE VALUE OF DELAY TIME PERIOD ≤ FIRST THRESHOLD?
N
Y
S506
SET DETECTED SIGNAL AS REFERENCE SIGNAL AND STORE REFERENCE TIME
END

FIG.5

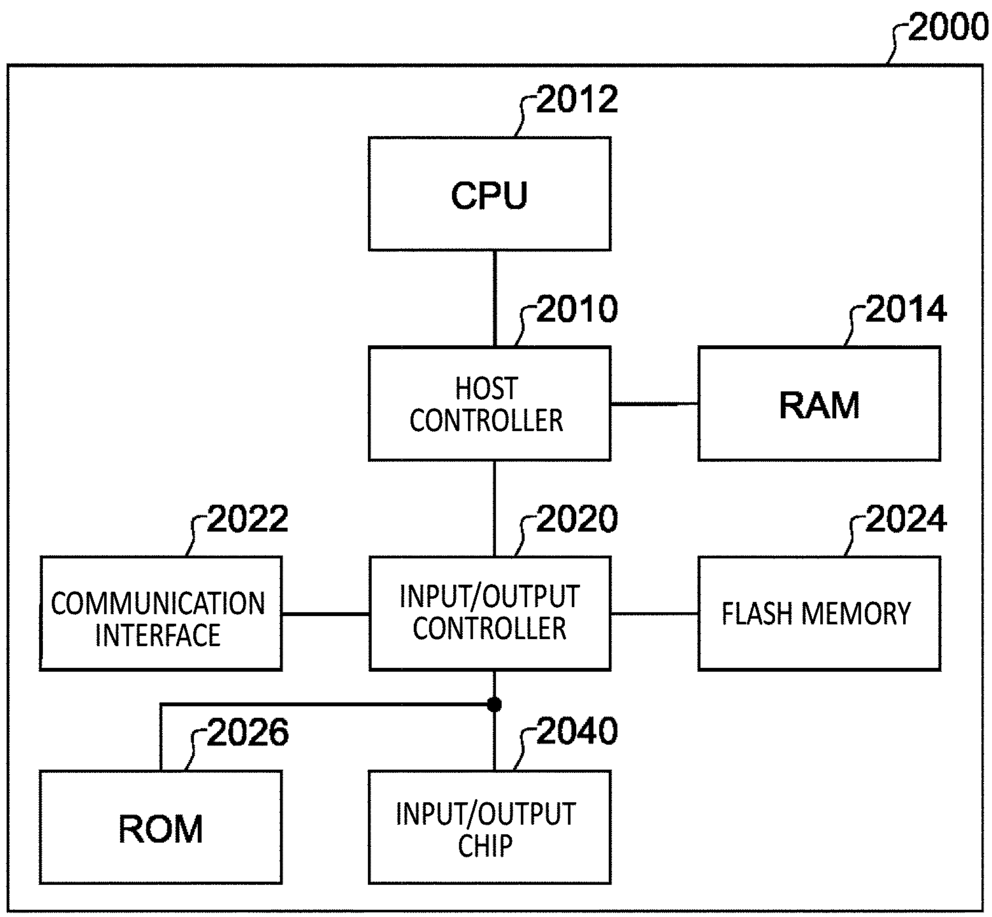
2000
2012
CPU
2010
HOST CONTROLLER
2014
RAM
2022
COMMUNICATION INTERFACE
2020
INPUT/OUTPUT CONTROLLER
2024
FLASH MEMORY
2026
ROM
2040
INPUT/OUTPUT CHIP

FIG.6

SIGNAL DETECTION APPARATUS, VEHICLE, AND METHOD

The contents of the following Japanese patent application(s) are incorporated herein by reference:

NO. 2022-107261 filed in JP on Jul. 1, 2022.

BACKGROUND

1. Technical Field

The present invention relates to a signal detection apparatus, a vehicle, and a method.

2. Related Art

Patent Document 1 and Patent Document 2 disclose techniques of detecting an illegal signal that is input to a communication network.

LIST OF CITED REFERENCES

Patent Document 1: Japanese Patent Application Publication No. 2021-136631

Patent Document 2: Japanese Patent Application Publication No. 2021-064921

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** conceptually illustrates a system configuration of a vehicle **10** in an embodiment.

FIG. **2** is a block diagram schematically illustrating a functional configuration included in an ECU **100**.

FIG. **3** is a diagram for describing processing of setting a reference time used to detect an illegal signal.

FIG. **4** is a diagram for describing processing for a determination unit **220** to determine an illegal signal.

FIG. **5** is a flowchart related to a signal detection method performed by an ECU **110**.

FIG. **6** illustrates an example of a computer **2000**.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

Hereinafter, the present invention will be described by way of embodiments of the invention, but the following embodiments are not for limiting the invention according to the claims. In addition, not all of the combinations of features described in the embodiments are essential to the solution of the invention.

FIG. **1** conceptually illustrates a system configuration of a vehicle **10** in an embodiment. The vehicle **10** includes a system **20**. The system **20** includes a plurality of electronic control units (ECUs) including an ECU **100**, an ECU **110**, an ECU **111**, an ECU **120**, and an ECU **121**. The ECUs included in the vehicle **10** include an ECU which controls equipment which directly affects travelling of the vehicle **10**, such as, for example, an engine, a gearbox, or a steering gear. The ECUs included in the vehicle **10** include an ECU which controls equipment which does not directly affect travelling of the vehicle **10**, such as, for example, an air conditioner or a navigation device. The ECU **100**, the ECU **110**, the ECU **111**, the ECU **120**, and the ECU **121** are examples of in-vehicle equipment.

The ECUs included in the vehicle **10** mutually perform communication by controller area network (CAN) communication. Each of the ECUs included in the vehicle **10** is connected so as to be mutually communicable by a plurality of CAN communication networks **180**. The ECU **100** functions as a gateway which relays communication between the plurality of CAN communication networks **180**. The CAN communication network **180** is an example of a communication network.

FIG. **2** is a block diagram schematically illustrating a functional configuration included in the ECU **100**. The ECU **100** includes a processing unit **200** and a storage unit **280**. The ECU **100** performs processing of determining whether there is a transmission of an illegal signal to the CAN communication network **180** by a third party impersonating an ECU included in the vehicle **10**, which is so-called an impersonation attack.

The processing unit **200** may be implemented by a processor such as a central processing unit (CPU) which performs computation processing. The storage unit **280** may include a nonvolatile storage medium such as a flash memory or a volatile storage medium such as a random access memory. The ECU **100** may be configured to include a computer. The ECU **100** performs various types of control by the processing unit **200** being operated in accordance with a program stored in the nonvolatile storage medium.

The processing unit **200** includes a signal detection unit **210** and a determination unit **220**. The signal detection unit **210** detects signals expected to be input to the CAN communication network **180** in a predetermined cycle. The determination unit **220** determines whether detection timing of a second signal detected by the signal detection unit **210** after a first signal which is set as a reference signal for determining an illegal signal is in a predetermined range within which reference timing that is timing after an integer multiple of a cycle from detection timing of the first signal falls among the signals detected by the signal detection unit **210**, and whether a shift between the detection timing of the second signal and the reference timing is a first threshold or less. When the shift between the detection timing of the second signal and the reference timing is the first threshold or less, the determination unit **220** sets the second signal as the reference signal. When the shift between the detection timing of the second signal and the reference timing exceeds the first threshold, the determination unit **220** sets the first signal as the reference signal without setting the second signal as the reference signal. In this manner, even when the second signal may be regarded as a normal signal, the determination unit **220** can set the first signal as the reference signal without setting the second signal as the reference signal when the detection timing is shifted from cycle timing, and can set the second signal as the reference signal when the detection timing is not shifted from the cycle timing. With this configuration, the shift of the reference signal due to a disturbance or the like can be avoided.

When the shift between the detection timing of the second signal and the reference timing is a second threshold or less, the determination unit **220** determines that the detection timing of the second signal is in the predetermined range within which the reference timing falls. The first threshold is less than the second threshold. With this configuration, while whether the signal is an illegal signal is accurately determined by setting the cycle timing as a reference, a normal signal hardly shifted from the cycle timing can be set as the reference signal.

The determination unit **220** determines that the second signal is a normal signal when the detection timing of the second signal is in the predetermined range. With this configuration, it is possible to accurately determine that the second signal is a normal signal.

When the detection timing of the second signal is not in the predetermined range, the determination unit **220** determines that the second signal is not a normal signal, and determines whether the second signal is an illegal signal based on information included in the second signal or/and another signal input to the CAN communication network **180**. With this configuration, it is possible to more accurately determine whether the signal is an illegal signal by using internal information included in the signal. That is, if the timings are detected by using a rear end of each signal (reception completion timing of a signal) as the reference, signal length information of the second signal itself which is included in a header of the second signal is used to judge whether the second signal is pushed to the outside of the range in relation to the first signal. To judge this more accurately, it is possible to determine from header information of the first signal or the like whether the second signal is a normal signal that is pushed to the outside of the range due to a relationship between the first signal and the second signal, or is an abnormal signal existing outside of the range due to other factors.

The detection timing of the first signal may be timing at which the signal detection unit **210** has completed the reception of the first signal, and the detection timing of the second signal may be timing at which the signal detection unit **210** has completed the reception of the second signal. For example, when the reception start timing of the signal (or transmission start timing from a transmission source) is used as the detection timing, to determine whether the second signal is delayed due to communication contention with a preceding signal or bus idle latency, a signal length of the preceding signal is to be taken into account in addition to the reception start timing of the preceding signal and the reception start timing of the second signal. That is, when the reception start timing of the signal is used as the detection timing, the signal length of the preceding signal is to be checked. On the other hand, when the reception completion timing of the signal (or transmission completion timing) is used as the detection timing, to determine whether the second signal is delayed due to communication contention with a preceding signal or bus idle latency, the signal length of the second signal is to be taken into account in addition to the reception completion timing of the preceding signal and the reception completion timing of the second signal, but the signal length of the second signal can be acquired from header information of the second signal. Therefore, by using the reception completion timing of the signal as the detection timing, the determination unit **220** can perform the determination from the internal information of the second signal itself without acquiring the signal length of the preceding signal.

FIG. **3** is a diagram for describing processing of setting a reference time used to detect an illegal signal. FIG. **3** illustrates a time at which the signal detection unit **210** has actually detected a signal on the CAN communication network **180**. Note that in the present embodiment, the determination unit **220** determines which one of a plurality of signals to which a same CAN ID has been added is a normal signal. Therefore, in the present embodiment, a description will be provided while focusing on signals to which a particular CAN ID has been added.

In FIG. **3**, an actual signal time is a time at which the signal detection unit **210** has actually detected a signal. In the present embodiment, the time at which the signal is detected may be a signal transmission start time. For example, the time at which the signal is detected may be a time at which the signal detection unit **210** has detected the transmission start of the signal. In another embodiment, the time at which the signal is detected may be a transmission completion time of the signal. For example, the time at which the signal is detected may be a time at which the signal detection unit **210** has detected the transmission completion of the signal.

In FIG. **3**, an actual signal cycle is a time interval between signals continuously input to the CAN communication network **180**. It is assumed that signals set as processing objects in the present embodiment are signals expected to be input to the CAN communication network **180** in a predetermined cycle of 10 ms. According to the present embodiment, to plainly illustrate the processing of setting the reference time, it is assumed that the first threshold is 0.5 ms.

It is assumed that a time t**1** is a time at which, after a state is established where communication through the CAN communication network **180** is allowed in the system **20**, the signal detection unit **210** receives a signal input to the CAN communication network **180** for the first time. At a time t**2** after an elapse of a time period of 11 ms from the time t**1**, the signal detection unit **210** detects a next signal. The determination unit **220** calculates, as a delay time period, a difference between the time t**2** and timing after an elapse of a time period equivalent to a cycle of 10 ms from the time t**1**. The delay time period at the time t**2** becomes 1 ms. Therefore, the determination unit **220** calculates 1 ms as a cumulative value of the delay time period at the time t**2**. Since the cumulative value of the delay time period exceeds the first threshold, the signal detection unit **210** does not set the signal received at the time t**2** as the reference signal.

Subsequently, the signal detection unit **210** further detects a next signal at a time t**3** after a further elapse of a time period of 11 ms from the time t**2**. The determination unit **220** calculates, as the delay time period, a difference between the time t**3** and timing after an elapse of a time period of the cycle of 10 ms from the time t**2**. The delay time period at the time t**3** becomes 1 ms. Therefore, the determination unit **220** calculates 2 ms as the cumulative value of the delay time period at the time t**3**. Since the cumulative value of the delay time period exceeds the first threshold, the signal detection unit **210** does not set the signal received at the time t**3** as the reference signal.

Subsequently, the signal detection unit **210** further detects a next signal at a time t**4** after a further elapse of a time period of 8 ms from the time t**3**. The determination unit **220** calculates, as the delay time period, a difference between the time t**4** and timing after an elapse of a time period of the cycle of 10 ms from the time t**3**. The delay time period at the time t**4** becomes −2 ms. Therefore, the determination unit **220** calculates 0 ms as the cumulative value of the delay time period at the time t**4**. Since the cumulative value of the delay time period is the first threshold or less, the signal detection unit **210** sets the signal received at the time t**4** as the reference signal, and sets the time t**4** as the reference time.

Subsequently, the signal detection unit **210** further detects a next signal at a time t**5** after a further elapse of a time period of 11.02 ms from the time t**4**. The determination unit **220** calculates, as the delay time period, a difference between the time t**5** and timing after an elapse of a time period of the cycle of 10 ms from the time t**4**. The delay time period at the time t**5** becomes 1.02 ms. Therefore, the determination unit **220** calculates 1.02 ms as the cumulative value of the delay time period at the time t**5**. Since the cumulative value of the delay time period exceeds the first threshold, the signal detection unit **210** does not set the signal received at the time t**5** as the reference signal.

Subsequently, the signal detection unit **210** further detects a next signal at a time t**6** after a further elapse of a time period of 8.5 ms from the time t**5**. The determination unit **220** calculates, as the delay time period, a difference between the time t**6** and timing after an elapse of a time period of the cycle of 10 ms from the time t**5**. The delay time period at the time t**6** becomes −1.5 ms. Therefore, the determination unit **220** calculates −0.48 ms as the cumulative value of the delay time period at the time t**6**. Since an absolute value of the cumulative value of the delay time period is the first threshold or less, the signal detection unit **210** sets the signal received at the time t**6** as the reference signal, and sets the time t**6** as the reference time. At this time, the determination unit **220** resets a cumulative value of the delay time period to 0.

Subsequently, the signal detection unit **210** further detects a next signal at a time t**7** after a further elapse of a time period of 10.4 ms from the time t**6**. The determination unit **220** calculates, as the delay time period, a difference between the time t**7** and timing after an elapse of a time period of the cycle of 10 ms from the time t**6**. Since the cumulative value of the delay time period is reset to 0 at the time t**6**, the delay time period at the time t**7** becomes 0.4 ms. Therefore, the determination unit **220** calculates 0.4 ms as the cumulative value of the delay time period at the time t**7**. Since the cumulative value of the delay time period is the first threshold or less, the signal detection unit **210** sets the signal received at the time t**7** as the reference signal, and sets the time t**7** as the reference time.

In this manner, when a new signal is detected, if a shift between the timing at which the new signal is detected and the reference timing that is timing after an integer multiple of the signal cycle from the reference time is the first threshold or less, the determination unit **220** sets the new signal as the reference signal, and sets a time at which the new signal is detected as the reference time used to detect an illegal signal.

FIG. **4** is a diagram for describing processing for a determination unit **220** to determine an illegal signal. With reference to FIG. **4**, a case will be described where a determination is made on whether a signal which is detected after a time t**11** is set as the reference time is an illegal signal. According to the present embodiment, to plainly illustrate the processing of determining whether the signal is an illegal signal, it is assumed that the second threshold for determining whether the signal is a normal signal is 1 ms. That is, it is assumed that when the time at which the signal is detected is in a range of ±1 ms in which the reference timing calculated from the reference time is set as a center, it is determined that the detected signal is a normal signal. According to the present embodiment, for the purpose of ease of illustration, a description will be provided where the first threshold (0.5 ms) is set as ½ of the second threshold (1 ms), but the first threshold may be ⅕ or less of the second threshold. The first threshold may be 1/10 of the second threshold.

With reference to FIG. **4**, the signal detection unit **210** detects a new signal at a time t**12** after an elapse of 11 ms from the time t**11**. The determination unit **220** sets a time after 10 ms from the reference time as the reference timing. A difference between the time t**12** and the reference timing is 1 ms. That is, the time t**12** is in the range of ±1 ms in which the reference timing is set as the center. Therefore, the determination unit **220** determines that the signal detected at the time t**12** is a normal signal.

Subsequently, the signal detection unit **210** detects a new signal at a time t**13** after an elapse of 10.5 ms from the time t**12**. The determination unit **220** sets a time after 10×2 ms from the reference time as the reference timing. A difference between the time t**13** and the reference timing is 1.5 ms. That is, the time t**13** is not in the range of ±1 ms in which the reference timing is set as the center. Therefore, the determination unit **220** determines that the signal detected at the time t**13** may be an illegal signal.

The determination unit **220** may determine whether the signal detected at the time t**13** is an illegal signal based on information included in the signal detected at the time t**13** and/or information included in the signal transmitted to the CAN communication network **180** immediately before the signal detected at the time t**13**. For example, the determination unit **220** may determine whether the signal detected at the time t**13** is delayed due to contention with another signal at the time of transmission based on a signal length included in header information of each signal and a detection time of each signal. When it is determined that the signal detected at the time t**13** is delayed due to contention with another signal at the time of transmission, the determination unit **220** may determine that the signal detected at the time t**13** is a normal signal. When it is determined that the signal detected at the time t**13** is not delayed due to contention with another signal at the time of transmission, the determination unit **220** may determine that the signal detected at the time t**13** is an illegal signal.

Subsequently, the signal detection unit **210** detects a new signal at a time t**14** after an elapse of 9.5 ms from the time t**13**. The determination unit **220** sets a time after 10×3 ms from the reference time as the reference timing. A difference between the time t**14** and the reference timing is 1 ms. That is, the time t**14** is in the range of ±1 ms in which the reference timing is set as the center. Therefore, the determination unit **220** determines that the signal detected at the time t**14** is a normal signal.

Subsequently, the signal detection unit **210** detects a new signal at a time t**15** after an elapse of 9.8 ms from the time t**14**. The determination unit **220** sets a time after 10×4 ms from the reference time as the reference timing. A difference between the time t**15** and the reference timing is 0.8 ms. That is, the time t**15** is in the range of ±1 ms in which the reference timing is set as the center. Therefore, the determination unit **220** determines that the signal detected at the time t**15** is a normal signal.

In this manner, when a new signal is detected, the determination unit **220** sets, as the reference timing, timing after an integer multiple of the cycle from the reference time. When the detection timing of the new signal is in a predetermined range within which the reference timing falls, the determination unit **220** may determine that the new signal is a normal signal. On the other hand, when the detection timing of the new signal is not in the predetermined range within which the reference timing falls, the determination unit **220** may determine that the new signal is an illegal signal.

FIG. **5** is a flowchart related to a signal detection method performed by the ECU **110**. FIG. **5** is a flowchart related to processing of setting a reference time. The processing of the flowchart in FIG. **5** may be performed each time the signal detection unit **210** detects a signal.

When the signal detection unit **210** detects a signal, in S**502**, the determination unit **220** calculates a cumulative value of the delay time period for the detect signal based on a currently set reference time, a current cumulative value of the delay time period, and a detection time of the signal. In S**504**, the determination unit **220** determines an absolute value of the cumulative value of the delay time period is the first threshold or less. When the absolute value of the cumulative value of the delay time period is the first threshold or less, in S**506**, the determination unit **220** sets, as the reference signal, a signal detected by the signal detection unit **210**, and the storage unit **280** stores a detection time of the signal as the current reference time.

In S**506**, when the absolute value of the cumulative value of the delay time period exceeds the first threshold, the determination unit **220** ends the processing of the present flowchart without updating the reference time.

Moreover, when the absolute value of the cumulative value of the delay time period exceeds the first threshold, under a condition that an elapsed period of time from the current reference time until the detection time of the signal exceeds a predetermined period of time, the determination unit **220** may set the detection time of the signal as the reference time. When a certain error exists in a cycle for a signal to be actually input to the CAN communication network **180**, an error is also caused in the reference timing calculated based on the reference time and the cycle. Therefore, when the reference time is not updated for a long period of time, since an error caused in the reference timing is accumulated, an erroneous determination on whether the signal is a normal signal may be performed. When an elapsed period of time from the current reference time exceeds a predetermined period of time, by setting the detection time of the new signal as the reference time, it may be possible to reduce a likelihood that the erroneous determination on whether the signal is a normal signal is performed due to the cumulative error.

In the CAN communication network **180**, when contention is caused at the time of input of a signal to the CAN communication network **180**, the signal is to be transmitted according to a priority rank by way of communication arbitration, and a delay may be caused until the signal is actually transmitted. Therefore, when the determination on an illegal signal is performed while a detection time of the delayed signal is set as the reference time, whether the signal is a normal signal may be erroneously determined. In contrast to this, according to the system **20**, when the signal detection unit **210** detects a new signal, if a shift between the timing at which the new signal is detected and the reference timing that is timing after an integer multiple of the signal cycle from the reference time is the first threshold or less, the determination unit **220** sets the new signal as the reference signal, and sets a time at which the new signal is detected as the reference time. With this configuration, it becomes possible to set the reference time more appropriately for detecting a signal which is input in an unauthorized manner to the CAN communication network **180**. Furthermore, it becomes possible to appropriately detect the signal which is input in an unauthorized manner to the CAN communication network **180**.

FIG. **6** illustrates an exemplary computer **2000** in which a plurality of embodiments of the present invention may be wholly or partially embodied. A program installed in the computer **2000** can allow the computer **2000** to: function as systems such as the system **20** according to embodiments or each unit of the systems, or as apparatuses such as the ECU **110** or each unit of the apparatuses; perform operations associated with the systems or each unit of the systems or with the apparatuses or each unit of the apparatuses; and/or perform processes according to embodiments or steps in the processes. Such a program may be executed by a CPU **2012** in order to cause the computer **2000** to execute a specific operation associated with some or all of the processing procedures and the blocks in the block diagrams described herein.

The computer **2000** according to the present embodiment includes the CPU **2012** and a RAM **2014**, which are mutually connected by a host controller **2010**. The computer **2000** also includes a ROM **2026**, a flash memory **2024**, a communication interface **2022**, and an input/output chip **2040**. The ROM **2026**, the flash memory **2024**, the communication interface **2022**, and the input/output chip **2040** are connected to the host controller **2010** via an input/output controller **2020**.

The CPU **2012** operates according to programs stored in the ROM **2026** and the RAM **2014**, and thereby controls each unit.

The communication interface **2022** communicates with other electronic devices via a network. The flash memory **2024** stores a program and data used by the CPU **2012** in the computer **2000**. The ROM **2026** stores a boot program or the like executed by the computer **2000** during activation, and/or a program depending on hardware of the computer **2000**. The input/output chip **2040** may also connect various input/output units such as a keyboard, a mouse, and a monitor, to the input/output controller **2020** via input/output ports such as a serial port, a parallel port, a keyboard port, a mouse port, a monitor port, a USB port, or an HDMI (registered trademark) port.

A program is provided via a network or a computer-readable storage medium such as a CD-ROM, a DVD-ROM, or a memory card. The RAM **2014**, the ROM **2026**, or the flash memory **2024** is an example of the computer-readable storage medium. The program is installed in the flash memory **2024**, the RAM **2014**, or the ROM **2026** and executed by the CPU **2012**. Information processing written in these programs is read by the computer **2000**, and provides cooperation between the programs and the various types of hardware resources described above. A device or a method may be actualized by executing operations or processing of information depending on a use of the computer **2000**.

For example, when communication is executed between the computer **2000** and an external device, the CPU **2012** may execute a communication program loaded in the RAM **2014**, and instruct the communication interface **2022** to execute communication processing based on processing written in the communication program. Under the control of the CPU **2012**, the communication interface **2022** reads transmission data stored in a transmission buffer processing region provided in a recording medium such as the RAM **2014** or the flash memory **2024**, transmits the read transmission data to the network, and writes reception data received from the network into a reception buffer processing region or the like provided on the recording medium.

In addition, the CPU **2012** may cause all or a necessary portion of a file or a database stored in a recording medium such as the flash memory **2024** to be read into the RAM **2014**, and execute various types of processing on the data on the RAM **2014**. Next, the CPU **2012** writes back the processed data into the recording medium.

Various types of information such as various types of programs, data, a table, and a database may be stored in the recording medium and may be subjected to information processing. The CPU **2012** may execute, on the data read from the RAM **2014**, various types of processing including various types of operations, information processing, conditional judgement, conditional branching, unconditional branching, information retrieval/replacement, or the like described in this specification and specified by instruction sequences of the programs, and write back a result into the RAM **2014**. In addition, the CPU **2012** may search for information in a file, a database, or the like in the recording medium. For example, when multiple entries, each having an attribute value of a first attribute associated with an attribute value of a second attribute, is stored in the recording medium, the CPU **2012** may search for an entry having a designated attribute value of the first attribute that matches a condition from the multiple entries, and read the attribute value of the second attribute stored in the entry, thereby obtaining the attribute value of the second attribute associated with the first attribute that satisfies a predefined condition.

The programs or software modules explained above may be stored in the computer-readable storage medium on the computer **2000** or in the vicinity of the computer **2000**. A recording medium such as a hard disk or a RAM provided in a server system connected to a dedicated communication network or the Internet can be used as the computer-readable storage medium. A program stored in the computer-readable storage medium may be provided to the computer **2000** via a network.

A program, which is installed on the computer **2000** and causes the computer **2000** to function as the ECU **110**, may work on the CPU **2012** or the like to cause the computer **2000** to function as each unit of the ECU **110**. The information processing written in these programs are read by the computer **2000** to cause the computer to function as each unit of the ECU **110**, which is specific means realized by the cooperation of software and the various types of hardware resources described above. Then, by the specific means realizing calculation or processing of information according to a purpose of use of the computer **2000** in the present embodiment, the unique ECU **110** according to the purpose of use is constructed.

Various embodiments have been explained with reference to the block diagrams and the like. In the block diagrams, each block may represent (1) a stage of a process in which an operation is executed, or (2) each unit of the device having a role in executing the operation. A specific stage and each unit may be implemented by a dedicated circuit, a programmable circuit supplied with computer-readable instructions stored on a computer-readable storage medium, and/or a processor supplied with computer-readable instructions stored on a computer-readable storage medium. The dedicated circuit may include a digital and/or analog hardware circuit, or may include an integrated circuit (IC) and/or a discrete circuit. The programmable circuit may include a reconfigurable hardware circuit including logical AND, logical OR, logical XOR, logical NAND, logical NOR, and other logical operations, and a memory element such as a flip-flop, a register, a field programmable gate array (FPGA), a programmable logic array (PLA), or the like.

The computer-readable storage medium may include any tangible device capable of storing instructions to be executed by an appropriate device. Thereby, the computer-readable storage medium having instructions stored therein forms at least a part of a product including instructions which can be executed to provide means which executes processing procedures or operations specified in the block diagrams. An example of the computer-readable storage medium may include an electronic storage medium, a magnetic storage medium, an optical storage medium, an electromagnetic storage medium, or a semiconductor storage medium. More specific examples of the computer-readable storage medium may include a floppy disk, a diskette, a hard disk, a random access memory (RAM), a read only memory (ROM), an erasable programmable read only memory (EPROM or flash memory), an electrically erasable programmable read only memory (EEPROM), a static random access memory (SRAM), a compact disk read only memory (CD-ROM), a digital versatile disk (DVD), a Blu-ray (registered trademark) disk, a memory stick, an integrated circuit card, or the like.

The computer-readable instruction may include an assembler instruction, an instruction-set-architecture (ISA) instruction, a machine instruction, a machine dependent instruction, a microcode, a firmware instruction, state-setting data, or either of source code or object code written in any combination of one or more programming languages including an object oriented programming language such as Smalltalk (registered trademark), JAVA (registered trademark), and C++, and a conventional procedural programming language such as a "C" programming language or a similar programming language.

Computer-readable instructions may be provided to a processor of a general purpose computer, a special purpose computer, or other programmable data processing device, or to programmable circuit, locally or via a local area network (LAN), wide area network (WAN) such as the Internet, and a computer-readable instruction may be executed to provide means which executes operations specified in the explained processing procedures or block diagrams. Examples of the processor include a computer processor, a processing unit, a microprocessor, a digital signal processor, a controller, a microcontroller, and the like.

While the present invention has been described by way of the embodiments, the technical scope of the present invention is not limited to the above described embodiments. It is apparent to persons skilled in the art that various alterations or improvements can be made to the above described embodiments. It is also apparent from the description of the claims that the embodiments to which such alterations or improvements are made can be included in the technical scope of the present invention.

The operations, procedures, steps, and stages etc. of each process performed by a device, system, program, and method shown in the claims, specification, or diagrams can be executed in any order as long as the order is not indicated by "before", "prior to", or the like and as long as the output from a previous process is not used in a later process. Even if the process flow is described using phrases such as "first" or "next" in the claims, specification, or drawings, it does not necessarily mean that the process must be performed in this order.

EXPLANATION OF REFERENCES

10: vehicle;
20 system;
100: ECU;
110: ECU;
111: ECU;
120: ECU;
121: ECU;
180: CAN communication network;
200: processing unit;
210: signal detection unit;
220: determination unit;
280: storage unit;
2000: computer;
2010: host controller;
2012: CPU;

2014: RAM;
2020: input/output controller;
2022: communication interface;
2024: flash memory;
2026: ROM;
2040: input/output chip.

What is claimed is:

1. A signal detection apparatus comprising:

a signal detection unit which detects signals expected to be input to a communication network in a predetermined cycle; and

a determination unit which determines whether detection timing of a second signal detected by the signal detection unit after a first signal which is set as a reference signal for determining an illegal signal is in a predetermined range within which reference timing that is timing after an integer multiple of the cycle from detection timing of the first signal falls among the signals detected by the signal detection unit, and whether a shift between the detection timing of the second signal and the reference timing is a first threshold or less, wherein

the determination unit

sets the second signal as the reference signal when the shift between the detection timing of the second signal and the reference timing is the first threshold or less,

sets the first signal as the reference signal without setting the second signal as the reference signal when the shift between the detection timing of the second signal and the reference timing exceeds the first threshold,

determines that the detection timing of the second signal is in the predetermined range within which the reference timing falls when the shift between the detection timing of the second signal and the reference timing is a second threshold or less, wherein the first threshold is less than the second threshold,

determines that the second signal is a normal signal when the detection timing of the second signal is in the predetermined range, and

determines that the second signal is not a normal signal when the detection timing of the second signal is not in the predetermined range, and determines whether the second signal is an illegal signal based on information included in the second signal or/and another signal input to the communication network.

2. The signal detection apparatus according to claim **1**, wherein

the determination unit determines that the second signal is not a normal signal when the detection timing of the second signal is not in the predetermined range, and determines whether the second signal is an illegal signal based on information included in the second signal and another signal input to the communication network.

3. The signal detection apparatus according to claim **1**, wherein

the detection timing of the first signal is timing at which the signal detection unit completes reception of the first signal, and the detection timing of the second signal is timing at which the signal detection unit completes reception of the second signal.

4. The signal detection apparatus according to claim **1**, wherein

the communication network is a controller area network (CAN) communication network.

5. The signal detection apparatus according to claim **4**, wherein

the first signal and the second signal are signals including a same CAN ID (Identifier).

6. The signal detection apparatus according to claim **1**, wherein the signal detection apparatus is installed in a vehicle.

7. A method comprising:

detecting signals expected to be input to a communication network in a predetermined cycle;

determining whether detection timing of a second signal detected after a first signal which is set as a reference signal for determining an illegal signal is in a predetermined range within which reference timing that is timing after an integer multiple of the cycle from detection timing of the first signal falls among the signals detected in the detecting, and whether a shift between the detection timing of the second signal and the reference timing is a first threshold or less;

setting the second signal as the reference signal when the shift between the detection timing of the second signal and the reference timing is the first threshold or less; and

setting the first signal as the reference signal without setting the second signal as the reference signal when the shift between the detection timing of the second signal and the reference timing exceeds the first threshold;

determining that the detection timing of the second signal is in the predetermined range within which the reference timing falls when the shift between the detection timing of the second signal and the reference timing is a second threshold or less, wherein the first threshold is less than the second threshold;

determining that the second signal is a normal signal when the detection timing of the second signal is in the predetermined range; and

determining that the second signal is not a normal signal when the detection timing of the second signal is not in the predetermined range, and determining whether the second signal is an illegal signal based on information included in the second signal or/and another signal input to the communication network.

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